Annual Reviews in Control 57 (2024) 100951

Contents lists available at ScienceDirect

Annual Reviews in Control

journal homepage: www.elsevier.com/locate/arcontrol

Broad-deep network-based fuzzy emotional inference model with personal information for intention understanding in human–robot interaction

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ARTICLE INFO

Keywords:
Human–robot interaction
Emotional intention
Broad learning
Convolution neural networks

ABSTRACT

A broad-deep fusion network-based fuzzy emotional inference model with personal information (BDFEI) is proposed for emotional intention understanding in human–robot interaction. It aims to understand students' intentions in the university teaching scene. Initially, we employ convolution and maximum pooling for feature extraction. Subsequently, we apply the ridge regression algorithm for emotional behavior recognition, which effectively mitigates the impact of complex network structures and slow network updates often associated with deep learning. Moreover, we utilize multivariate analysis of variance to identify the key personal information factors influencing intentions and calculate their influence coefficients. Finally, a fuzzy inference method is employed to gain a comprehensive understanding of intentions. Our experimental results demonstrate the effectiveness of the BDFEI model. When compared to existing models, namely FDNNSA, ResNet-101+GFK, and HCFS, the BDFEI model achieved superior accuracy on the FABO database, surpassing them by 12.21%, 1.89%, and 0.78%, respectively. Furthermore, our self-built database experiments yielded an impressive 82.00% accuracy in intention understanding, confirming the efficacy of our emotional intention inference model.

1. Introduction

Emotion understanding involves examining the connection between outward actions and emotional states, and is considered a cognitive process (Kazemifard et al., 2014). In cognitive psychology, intentions refer to the thought or plan of a person's cognition prior to an action (Hersh, 2013). Every intention produced by a person is closely related to emotion, and every emotional state is a reaction and facial emotion of his inner intention (Chen et al., 2020d). When it comes to emotional communication (Picard, 1997), facial emotions and gestures are particularly potent, capable of transmitting as much as 70% of information. Intelligent robotics is advancing rapidly (Sun et al., 2022), but there are still few robots capable of understanding human emotions and intentions (Li & Deng, 2022). Therefore, the utilization of visual information, such as facial emotions and gestures, in human–robot interaction can lead to a more accurate interpretation of human emotional intentions.

Students' facial emotions and behaviors in class can truly reflect their emotional state, providing a basis for evaluating teaching effectiveness. Currently, many studies on teaching effectiveness evaluation rely on subjective data such as questionnaire surveys (Liu et al., 2022). However, we need more objective data to analyze various methods, which will have greater objectivity and allow us to further explore the accuracy of results obtained from subjective data (Li et al., 2023a). Hence, it is imperative to gather more unbiased information about students' learning progress during class. Affective computing research primarily concentrates on recognizing emotions through various modalities, including facial emotions, body gestures, speech, physiological signals, and other forms of multimodal information (Al Chanti & Caplier, 2021).

The development of a multimodal emotion feature set and multimodal fusion is a crucial aspect of understanding emotional intentions across multiple modes (Schuller et al., 2011). Although they are all visual information, facial emotion and body gesture feature extraction algorithms are different due to different modes. Facial feature

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https://doi.org/10.1016/j.arcontrol.2024.100951
Received 5 November 2023; Accepted 15 March 2024
Available online 23 March 2024
1367-5788/

extraction can be categorized into temporal and spatial sequence-based algorithms and algorithms based on static images (Chen et al., 2022). The extraction methods of gesture feature mainly include the extraction methods based on global feature and local feature (Chen et al., 2013). The majority of research has concentrated on machine learning based on visual information. Principal component analysis is often employed to extract emotion feature data due to the high dimensionality and data redundancy of emotion images (Zeng et al., 2022). The use of deep learning (Lee et al., 2020a) has become increasingly prevalent in the field of emotion recognition, thanks to its ability to extract abstract and sophisticated features from emotional image data (Liu et al., 2020a). In recent years, various deep learning networks, including recurrent neural networks and convolutional neural networks (CNN) (Saviolo & Loianno, 2023; Thuseethan et al., 2023), have been utilized for this purpose (Li et al., 2023b).

Deep learning is highly versatile when it comes to emotion recognition (Rouast et al., 2021). For instance, a deep system proposed for recognizing emotions from facial expressions using edge computing (Muhammad & Hossain, 2021). Based on singular value decomposition-based co-clustering a novel feature selection strategy was introduced (Khan et al., 2020) to identify the most important facial features. In another study (Farajzadeh & Hashemzadeh, 2018), an exemplar-based SVM method was developed, which combined SVM with a histogram of oriented gradients. Despite the effectiveness of these methods in emotion recognition, the challenge of how to apply the results to real-world scenarios remains unresolved (Bruns, 2006).

Our proposal's primary innovation stems from the fusion of broad learning and deep learning techniques in emotion recognition, coupled with the distinctive fuzzy reasoning model that incorporates personal information to enhance intention understanding. The main contribution of our proposal are as follows. First, to understand coexistence bimodal information, the features of depth and width dimensions is obtained by broad-deep fusion network. What is more, multi-factor analysis method is employed to construct a knowledge model to select the key factors affecting emotional intention. Finally, a unique fuzzy reasoning model is applied, which integrates personal information, behavior and expression to understand the emotional intention of students in the teaching scene. The result of intention understanding and emotion recognition on FABO and the self-built database proves the effectiveness of our proposal.

2. Method overview

Emotional intention understanding is a cognitive process that focuses on how emotion and outward behavior interact and uses emotion to speculate an individual's emotion state (Coelho et al., 2017). How research on emotional understanding can be applied to real-world situations has become an issue of increasing interest to researchers. Facial expressions are a significant channel for conveying emotional information, accounting for 55% of the total information in diverse emotional communication forms, as highlighted by Chen et al. (2021). In the context of human–robot interaction, facial emotion recognition holds a pivotal role, as emphasized by Wu et al. (2021). It is worth noting that the expressions and behaviors of students during classroom interactions can serve as authentic indicators of their emotional states and can be used as the basis for evaluation of teaching effectiveness. Therefore, the realization of students' emotional intention understanding can provide more objective basis for teaching evaluation and effectively serve the teaching scene.

Machine learning enables systems for human–robot interaction to autonomously learn from original data. However, the representation of data is always a key component of existing machine learning algorithms (Yin et al., 2017). Moreover, establishing a sufficient set of feature representations is extremely challenging for many jobs involving human–robot interaction (Chen et al., 2023). The issue can be resolved using the representation learning method. In general, learning representations outperform the manual design (Shehu et al., 2022). However, it might be quite challenging to extract sophisticated and abstract elements (Chen et al., 2020a) from genuine emotional image data (Chen et al., 2021b). Deep learning (Lee et al., 2020b) gathers features from the entire image using weight sharing through convolution operation (Liu et al., 2020b), such as deep belief networks (DBNs) (Kong et al., 2022; Mohan et al., 2021; Xia & Liu, 2015), CNNs, and RNNs. While deep networks are commonly employed, it is important to acknowledge that their depth structure often encompasses a substantial number of parameters. As a result, parameter optimization requires a step-by-step update process using gradient descent method. The updating and even training of the network is a time-consuming process (Wang et al., 2021). Broad learning (Chen & Liu, 2017), established from random vector function-link neural networks, achieves feature representation by horizontally expanding the network, while CNN enhances feature representation capability through longitudinal network deepening.

Based on the above development, emotion recognition has been well realized, and then the problem of integrating emotion recognition into real scene application needs to be further studied. Fuzzy emotion reasoning model is a kind of emotion analysis method based on fuzzy logic, which can reason and analyze uncertain or fuzzy emotion information. The two-layer fuzzy SVR-TS model is proposed by Chen et al. (2020b) for dynamic emotion understanding. Similarly, an approach for evaluating distance education quality was introduced (Liu et al., 2022), which relies on multi-granularity probabilistic linguistic term sets and disappointment theory. on this basis, the BDFEI is proposed follows the steps mentioned above. Initially, we preprocess facial and behavioral images of students. Subsequently, we employ convolution and pooling, common techniques in deep learning, for feature extraction. The emotion recognition is implemented using the ridge regression algorithm. Finally, a fuzzy inference model is employed to identify the emotional intentions of students. Specific details regarding the experimental setup, database usage, and experimental design will be elaborated in the experimental section.

3. Emotional intention understanding using BDFEI

Due to the complexity of emotional feature and the ambiguity of intention, we adopted emotion feature classification based on broad-deep fusion network. Convolution and pooling are used for feature extraction and Broad learning is applied for emotional recognition. The obtained behavior, expression, and personal information are each independently represented as discrete label values. Then, fuzzy reasoning mainly realizes the integration of behavior, expression and personal information, so as to get students' intention in the teaching scene. Fig. 1 shows the structure of BDFEI for emotional intention understanding.

3.1. Emotion recognition based on CNN

The proposed facial emotion recognition method employs a convolutional neural network that includes two convolutional and pooling layers, a dropout layer, and a fully connected layer, with SoftMax used for classification. 40 convolution kernels of size 25*25 is used in the first convolution layer, while 40 convolution kernels of size 7*7 in the second convolution layer. Both layers use pooling kernels of size 2*2. The convolution layer performs the following calculations,

$$y_j^l = \theta\left(\sum_{i=1}^{N_j^{l-1}} \omega_j * x_i^{l-1} + b_j^l\right), j = 1, 2, \ldots, M, \tag{1}$$

Each feature map of convolution layer y_j^l is obtained by convolving the corresponding convolution kernel ω_j with the feature maps x_i^{l-1} of the previous layer, and adding a bias term b_j^l. The resulting values are then passed through the activation function $\theta()$. N_j^{l-1} represents

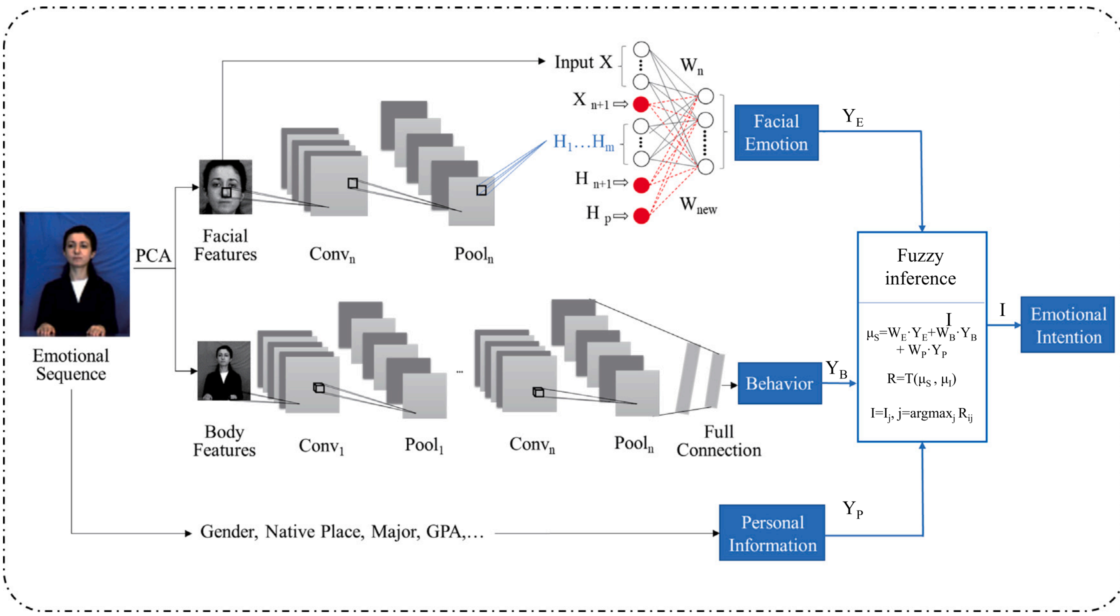

Fig. 1. The structure of BDFEI for emotional intention understanding.

the number of features for each feature map and M is the number of feature maps for each convolutional layer.

The pooling layer is employed for secondary feature screening to obtain more representative features. Different pooling algorithms, such as random pooling, average pooling, and maximum pooling, are commonly used in CNN depending on the requirements of the algorithms. In particular, one has

$$S_{i,j} = \sum_{i=1,j=1}^{c} p_{ij} x_{ij}, \tag{2}$$

After the pooling operation the feature map is $S_{i,j}$. p_{ij} is the parameter of convolution kernel of the ith convolution kernel in the jth layer, c is the size of the pooling window.

Overfitting is a common issue encountered in deep learning. To alleviate this problem, the dropout layer can be employed to improve feature learning efficiency and prevent overfitting. During the training phase of the CNN network, the dropout layer modifies the working state of neuron nodes, as demonstrated in (3). During the testing phase, the entire network is used to make predictions. Each neuron is activated and the output of each layer is calculated in a forward pass with (4).

$$DropoutTrain(x) = RandomZero(c) \times x \tag{3}$$

$$DropoutTest(x) = (1 - c) \times x \tag{4}$$

The SR layer serves as the output layer of the network, responsible for classifying emotion features. For training set $\{(x^{(1)}, y^{(1)}), \ldots, (x^{(m)}, y^{(m)})\}$, $y^{(i)} \in \{1, 2, \ldots, k\}$.

3.2. Behavior recognition based on broad-deep fusion network

Feature extraction is realized by convolutional and pooling layer. Different from the size of the expression images in Section 3.1, the kernel sizes selected is also different. The first and second convolution layers in the network consist of 40 and 50 convolution kernels respectively, with sizes of 15*25 and 6*7. Both layers also use 2*2 pooling kernels. Behavior feature classification is accomplished by applying the broadening ridge regression algorithm. In this method, the weights of the BL networks are stored in a pseudo-inverse matrix. Calculating the pseudo-inverse of feature nodes and enhancement nodes to the target value is a crucial step in the BL network. To establish a mapping from input data to feature nodes, the following steps are employed:

1. First, the matrix H_1' with dimensions $s \times f$ is normalized using the z-score method, and an additional column of 1 is appended to create a matrix H_1 with dimensions $s \times (f + 1)$.

2. Next, a random weight matrix w_e is generated from a Gaussian distribution. The elements of W_{e_i} are updated based on w_e, where i represents the current iteration, and N_2 is the total number of iterations.

3. Then, The random feature vector A_1 is computed as the product of H_1 and w_e, with sparsity enforced through a sparse autoencoder.

4. Finally, the feature node T_1 is derived as the result of normalizing the matrix product of H_1 and W using the operation denoted by *normal*.

Throughout this process, the training set H_1', with dimensions $s \times f$, is employed, where s denotes the number of samples, and f signifies the number of features. The quantity of feature nodes in each window is denoted as $N1$, and the number of windows is designated as N_2. The complete feature node matrix for the entire network is represented as y.

The enhancement nodes within the broad learning network serve as a random complement to the feature nodes. Initially, in the BL approach, the feature node matrix is normalized and extended to create H_2. However, in our proposed method, H_2 is substituted with the feature matrix after convolution and pooling to avoid information loss caused by the random mapping process. The coefficient matrix of the enhanced nodes is a normalized random matrix, distinct from that of the feature nodes.

Then, the matrix containing the active enhanced nodes is denoted as T_2.

$$T_2 = tansig \frac{H_2 \times w_h \times s}{max(H_2 \times w_h)}, \tag{5}$$

The scaling factor of the enhancement node is denoted as s, and the coefficient matrix of the enhanced node is represented as w_h. The inputs are normalized, and the active range of the *tansig* function is within $[-1, 1]$, so the *tansig* function is chosen here.

The final input T_3 is calculated as

$$T_3 = \begin{bmatrix} Y_1 \\ T2 \end{bmatrix}. \tag{6}$$

The weight matrix W can be obtained using the ridge regression algorithm (Chen & Liu, 2017) in the following manner:

$$argmin : \|T_3 W - Y\|_v^{\sigma_1} + \lambda \|W\|_u^{\sigma_2}, \tag{7}$$

Table 1
The status with different expressions and behaviors.

Status / Behavior / Expression	Listening	Not listening
Happiness	PV	NV
Neutral	PM	NV
Boredom	PN	NV
Puzzlement	NM	NV
Disgust	NV	NV
Uncertainty	NV	NV

Here, it is important to note that $\sigma_1 > 0, \sigma_2 > 0$, and σ_1, σ_2, u and v are standard regularization parameter. We apply l2 norm regularization with $\sigma_1 = \sigma_2 = v = u = 2$. The matrix Y represents the output.

The parameter λ plays a critical role in imposing constraints on the square weights of W within the ridge regression algorithm. Conversely, as λ approaches infinity, the solution of the network tends towards zero. Specifically, when $\lambda = 0$, the solution is derived from the least square problem.

$$argmin : W = (\lambda I + T_3 T_3^T)^{-1} T_3^T Y, \tag{8}$$

By applying this method, the updating of network's weights can be achieved by the pseudo-inverse calculating only for the recently added nodes.

3.3. Emotional intention understanding based on fuzzy inference

Fuzzy inference mainly realizes the integration of behavior, expression and personal information, so as to get students' intention in the teaching scene. Firstly, multi-factor analysis method is adopted to select the key factors affecting the intention in our previous work [(Chen et al., 2018)], and their importance are calculated and analyzed. Based on this analysis, the expert experience is obtained to establish fuzzy rules between status and two of the factors (behavior and expression), as shown in Table 1, and the fuzzy relationship is calculated. Finally, the intention is obtained through a fuzzy decision. The status of the students determines their intentions. There are five status of students, very positive, positive, average, negative, very negative, abbreviated as PV, PM, PN, NM, NV.

The five types of personal status of the student are represented by a fuzzy set S. The relationship between this status and expressions, behaviors, and personal information is expressed using Zadeh's notation as Eq. 9.

$$S = \sum_{i=1}^{M_1} \frac{\mu_S(X_i)}{X_i}, \tag{9}$$

where X_i represents factors influencing Students' personal status, including facial expressions, behaviors, and personal information (gender, native place, major and GPA), with M_1 being the number of these factors, $\mu_S(X_i)$ represents the membership degree of X_i to the fuzzy set S.

The expression obtained by the neural network for facial emotion labels is denoted as Y_E, for behavioral labels as Y_B, and for personal information as Y_P, the number of categories for them are N_E, N_B, N_P, respectively. Then the calculation of μ_S is as follows

$$\mu_S = w_E \cdot Y_E + w_B \cdot Y_B + w_P \cdot Y_P, \tag{10}$$

where $\cdot$ represents multiplication, w_E, w_B, w_P represent the weights for facial expression, behavior, and personal information, respectively, and these weights are equidistant distribution values in the range [0, 1]. The specific calculation for W is as follows

$$N_1 = max(N_E, N_B, N_P), \tag{11}$$

$$\Delta = 1/N_1, \tag{12}$$

$$W = \{W_i | W_i = 1/2 \pm \triangle \cdot i, i = 1, 2, 3, \dots, N_1/2\}, \tag{13}$$

When w_E, w_B, w_P are obtained respectively by Eq. 13, μ_S is calculated using Eq. 10. Due to the variations in the values of Y_E, Y_B, and Y_P for different students, μ_S also undergoes changes.

The intention of the student is represented by a fuzzy set I. The relationship between intention and S is expressed as Eq. 14.

$$I = \sum_{j=1}^{N_2} \frac{\mu_I(I_j)}{I_j}, \tag{14}$$

where I_j represents the jth emotional intention, encompassing categories such as listening, understanding, confusion, disdain, uncertainty, etc., with N_2 being the number of these categories, $\mu_I(I_j)$ represents the membership degree of I_j to the fuzzy set I. I represents emotional intentions. M_j denotes the number of training samples for the jth intention, and M_2 represents the total number of intentions. Then

$$\mu_{I_j} = M_j / M_2, \tag{15}$$

Then the fuzzy relation is

$$R_{ij} = T(\mu_S, \mu_I) = \mu_{S_i} \times \mu_{I_j}, \tag{16}$$

where T stands for the product t-norm, $\times$ denotes element-wise multiplication. Students' personal status S' can be roughly divided into five categories: very good, good, average, poor and very poor. According to the fuzzy relationship between intention and personal status calculated by Eq. 16, final intention I is inferred as

$$I = I_j, j = argmax_j R_{ij}, \tag{17}$$

Through the fuzzy inference model, the emotional intentions of students are obtained based on the input of facial expressions, behaviors, and personal information.

4. Experiments and results

Two experiments were carried out to assess the effectiveness of the proposed method. The first experiment involved bimodal emotion recognition using a broad-deep fusion network, while the second experiment focused on emotional intention understanding through BD-FEI with behavior recognition, expression recognition and personal information. Finally, the experimental results are analyzed.

4.1. Experiment settings

Fig. 2 shows the emotion social robot interaction system we built for multimodal emotion understanding. It mainly includes an emotional robot, routers, data transmission equipments, and computing workstations. The workstation is equipped with Intel i5-4590 CPU, a 4.00 GB RAM, 3.3 GHZ frequency and 64-bit operating system type. As the software of experiment, we chose MATLAB R2018a. The bimodal emotion database called FABO database was adopted for the experiment.

The bimodal facial and gesture database was first introduced in 2005 (*A bimodal face and body gesture database*, 2006) for the purpose of automatically analyzing human nonverbal emotional behavior. The database was created in a laboratory setting by providing instructions and requesting participants to perform specific movements and actions while their visual data on gestures was collected shown as Fig. 3. 281 videos that cover eleven different emotions are used.

The self-built database comprises nine 3-minute videos recorded during classroom learning, featuring facial emotions and behaviors of seven undergraduate students. Each video shows the status of 2–3 students, and we selected 3 frames of each student's images for the

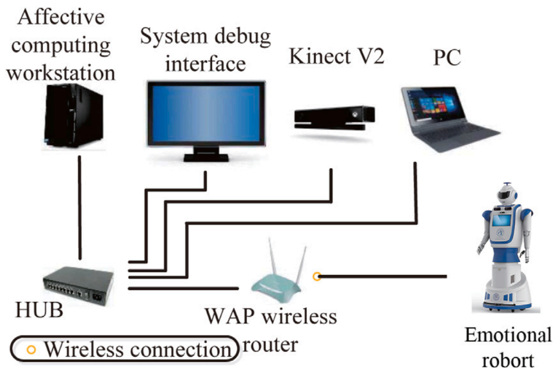

Fig. 2. The structure of emotion social robot interaction system.

Fig. 3. Sample images in FABO database.

Fig. 4. Sample images in self-built database.

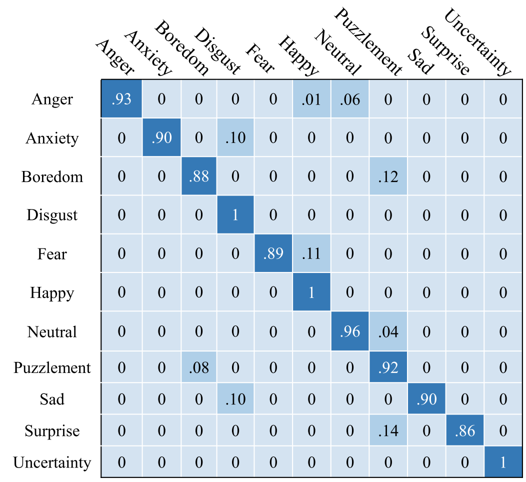

Fig. 5. Confusion matrix by BDFEI.

Table 2
Bimodal emotion recognition: comparative results.

Method changes	Accuracy(%)
FDNNSA (Chen et al., 2020c)	80.10
ResNet-101+GFK (Verma & Choudhary, 2021)	91.20
HCFS (Barros et al., 2018)	92.31
BDFEI	93.09

Table 3
The weight vector of each factor to the status.

Weight / Factor / Status	Expression	Behavior	Gender
PV	0.9	0.7	0.5
PM	0.7	0.7	0.5
PN	0.5	0.5	0.5
NM	0.3	0.3	0.5
NV	0.1	0.3	0.5

experiments. The videos were recorded from three different periods of the class. Fig. 4 displays sample images extracted from the database.

In the experiment, two databases are chosen, namely FABO database and the self-built database. Two experiments were conducted to verify the effectiveness of our proposal. The first is bimodal emotion recognition with broad-deep fusion network, and the second is emotional intention understanding based on BDFEI with behavior recognition, facial emotion recognition and personal information. Finally, the experimental results are analyzed.

4.2. Results on broad-deep fusion network for emotion recognition

The recognition for emotion status is an important prerequisite for emotional intention understanding, so the experiment of bimodal emotion recognition is firstly carried out based on broad-deep fusion network. Fig. 5 shows the bimodal emotion recognition results based on FABO database by confusion matrix.

The recognition accuracy of our two-modal emotion recognition method reaches 93.09%. and the recognition accuracy of disgust, happiness and uncertainty reached 100%. We compared our proposal with three other basic methods include a fuzzy deep neural network with sparse autoencoder (FDNNSA) (Chen et al., 2020c), a hierarchical classification fusion strategy for affect recognition (HCFS) (Barros et al., 2018), and residual Network combined with geodesic flow kernel (ResNet-101+GFK) (Verma & Choudhary, 2021). Table 2 shows the results of this comparison. Compared with the above three methods, the proposed method has higher recognition accuracy.

4.3. Results on BDFEI for intention understanding

In the intention understanding experiment, we divided the behavior into two parts based on the self-built database: listening to the teacher and not listening to the teacher. Listening to the teacher included happiness, neutral, boredom, puzzlement and disgust in the original database. The other emotional categories are not listening. Only when students are in the state of listening, the algorithm will recognize their expressions and infer their emotional intentions. Personal information is restricted by the database to include gender only.

There are five kinds of students' emotional intentions (listening, understanding, confusion, disdain and uncertainty). And three factors affect the status (behavior, expression and gender). The weight vector of each factor to the status are determined according to the previous statistical and analysis, as shown in Table 3.

The degrees of membership were artificially set as $I = 0.6/L + 0.4/U + 0.4/C + 0.2/D + 0.5/U$ for intention in the experiment. 10 images were selected for each of the 6 expressions in Table 2. Table 4 shows the result of intention understanding. Fig. 6 presents an example of recognition results for a test image obtained using the BDFEI method, including expressions, behaviors, personal status, and the emotional intentions.

According to Table 4, the accuracy of understanding emotional intention was 82.00%, indicating that our proposal can effectively

Table 4
The result of the emotional intention understanding.

Intention	Accuracy of intention understanding
Listening	18/20
Understanding	15/20
Confusion	13/20
Disdain	16/20
Uncertainty	20/20

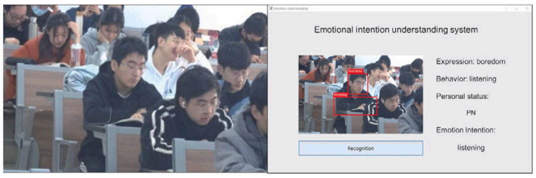

Fig. 6. A visual example of recognition results.

capture the emotional intention of students in the teaching scene. Among them, the recognition rate of ‘confusion’ is only 65%, and the part that was not recognized correctly is classified as ‘uncertainty’, while the recognition rate of ‘uncertainty’ is 100%. Overall, our proposal achieved better performance in both intention understanding and emotion recognition tasks.

5. Discussion

The proposed broad-deep fusion network-based fuzzy emotional inference model with personal information (BDFEI) achieves multimodal emotion recognition and emotional intention understanding in university teaching scenes.

Effectiveness of the BDFEI Model: The experimental results affirm the effectiveness of the BDFEI model. The model outperformed existing models, including FDNNSA (Chen et al., 2020c), ResNet-101+GFK (Verma & Choudhary, 2021), and HCFS (Barros et al., 2018), achieving a remarkable accuracy improvement on the FABO database. The substantial accuracy gains of 12.21%, 1.89%, and 0.78% over the aforementioned models showcase the superior performance of the BDFEI model in emotional intention understanding. In terms of emotional intention understanding, our recognition accuracy is 82.00%, and when the FDNNSA method (Chen et al., 2020c) is applied to our self-built database, its recognition rate is 78.00%. This suggests that the integration of the broad-deep fusion network, and fuzzy inference method has led to a more robust and accurate emotional intention understanding framework.

Feature Extraction and Emotional Behavior Recognition: The utilization of convolution and maximum pooling for feature extraction followed by the application of the ridge regression algorithm for emotional behavior recognition addresses the challenges associated with complex network structures and slow network updates in deep learning. This approach not only enhances computational efficiency but also contributes to the model’s ability to capture subtle emotional features in human–robot interactions.

Key Personal Information Factors: The incorporation of multivariate analysis of variance to identify key personal information factors influencing intentions is a novel aspect of the BDFEI model. The calculation of influence coefficients provides valuable insights into the relative importance of different personal information variables. This personalized approach enhances the adaptability of the model, allowing it to tailor emotional intention understanding to individual characteristics, which is particularly crucial in teaching scenes.

Fuzzy Inference for Comprehensive Understanding: The application of a fuzzy inference model as the final step in the model’s architecture contributes to a comprehensive understanding of intentions. Fuzzy logic allows for the modeling of uncertainty and imprecision in emotional states, enabling a more nuanced interpretation of human intentions. This aspect is particularly relevant in dynamic and varied university teaching scenes where emotions can exhibit a wide range of expressions.

Implications and Future Directions: The success of the BDFEI model in both the FABO database and the self-built database experiments underscores its potential for real-world applications in human–robot interaction. Considering the manually set parameters in the fuzzy inference process, we will explore adaptive optimization methods for model refinement and apply the model to more scenes in the future.

In conclusion, the BDFEI model, with its innovative integration of broad-deep fusion networks and personalized emotional inference model, presents a promising avenue for emotional intention understanding in human–robot interaction. The model’s superior performance on benchmark datasets and its adaptability to personal information factors highlight its significance in contributing to the field of affective computing. As an intention understanding model, it can be applied to various scenes such as customer service, healthcare, and virtual assistants. In the field of control systems, the BDFEI model’s understanding of emotions enhances system adaptability and responsiveness, opening up new possibilities for formulating more effective personalized control strategies. It also contributes to the development of control systems that align with user preferences and comfort. Furthermore, interdisciplinary research across affective computing, control theory, and human–robot interaction is expected to generate innovative applications that will advance the development of emotion-sensing control systems.

6. Conclusions

The BDFEI is proposed to facilitate the comprehension of emotional intentions in human–robot interaction. The random mapping at the original broad learning network is replaced by convolution and pooling layers for feature extraction and recognition. In addition, fuzzy inference is employed for intention understanding with expression, behavior and personal information. According to experiment results, our proposal outperforms on accuracy of emotion recognition than the methods mentioned above, such as ResNet-101+GFK, FDNNSA, etc. Moreover, our proposal can effectively capture the emotional intention of students in the teaching scene.

The rules of fuzzy affective reasoning model in this paper are artificial, so constructing a fuzzy reasoning model with learning capabilities is one of the directions for further research. In the future research, we will delve deeper into the exploration of explainable artificial intelligence (XAI) methods of emotional intention understanding, and complete the application experiment in real teaching scene.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work is supported by National Natural Science Foundation of China under Grants 61973286, 61603356, and 61733016, the 111 project under Grant B17040, and the Fundamental Research Funds for the Central Universities, China University of Geosciences (No. 2021063).

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